MailFlow AI Data Design

CSE 416 Fall 2026 | Data Design and Web Views | 20 October 2026

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Purpose and scope

This specification records the data consumed and produced by the current MailFlow implementation, the data stored by its SQLite layer, and the data used by the separate web-view preview. Gmail owns message content and mailbox state. MailFlow stores account identity and priority-classification metadata locally. The /demo route is an in-memory interface preview; its sample messages, drafts, and calendar values are not persisted or sent to Google services.

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1 System boundary

The live Next.js application authenticates a Google account, retrieves Gmail messages, derives a MailFlow priority, and returns message data to the browser. A separate SQLite database records local account identity, the latest priority per message, and priority transitions. The application does not maintain a local copy of Gmail message bodies, subjects, attachments, mailbox labels, or OAuth credentials.

The /demo route is the assignment web-view preview. It renders fixed sample records and handles interaction in React state. Its compose, review, send, and calendar controls are illustrative and do not call Gmail or Google Calendar.

Implementation: frontend/src/app/page.tsx; frontend/src/app/api/gmail/messages/route.ts; frontend/src/lib/mailflow-db.ts; frontend/src/app/demo/demo-client.tsx

2 Input and output data

2.1 Inputs to the live application

OAuth callback Google returns an authorization code and state. The callback compares the returned state with an HttpOnly cookie, exchanges the code for credentials, and stores those credentials in a process-local session map keyed by a random session ID.

Gmail account The Gmail profile supplies emailAddress for the local users row. The profile route may also supply a display name from Google userinfo; if unavailable, it uses the email local part as a fallback.

Gmail messages The list route consumes message IDs, pagination tokens, result counts, label IDs, internalDate, headers, snippet, thread ID, and MIME payload. Full message payloads are parsed for browser display. Existing MailFlow priority labels may determine the classification.

HTTP parameters after is a millisecond timestamp used for Daily Briefing. mode selects all, starred, drafts, or trash. pageToken selects a Gmail page for ordinary mailbox views.

User actions The message route accepts read, star, restore, and priority PATCH actions, plus DELETE for moving a message to Trash. Manual priority accepts urgent, fyi, or low.

Server configuration GOOGLE_CLIENT_ID, GOOGLE_CLIENT_SECRET, and GOOGLE_REDIRECT_URI configure OAuth. MAILFLOW_DB_PATH optionally overrides the SQLite file location.

Implementation: frontend/src/app/api/google/connect/route.ts; frontend/src/app/api/google/callback/route.ts; frontend/src/app/api/google/profile/route.ts; frontend/src/app/api/gmail/messages/route.ts; frontend/src/app/api/gmail/messages/[id]/route.ts

2.2 Outputs and destinations

Browser The list route returns connected, parsed message fields, priority, and, for ordinary mailbox views, nextPageToken and resultSizeEstimate. The parsed message includes an HTML document for an iframe, unread/starred/draft flags, subject, sender, date, snippet, message ID, and thread ID.

Gmail The application creates hidden MailFlow priority labels when absent; adds a classification label; replaces priority labels after manual changes; modifies STARRED or UNREAD; and calls trash or untrash for the corresponding actions.

SQLite The application upserts the user and writes current priority and priority-event records. Message content is not written to these tables.

HTTP response A missing session returns 401. Gmail route failures return an error code with status 502; invalid manual priority returns 400.

Implementation: frontend/src/lib/gmail-message.ts; frontend/src/lib/gmail-priority.ts; frontend/src/lib/mailflow-db.ts; frontend/src/app/api/gmail/messages/route.ts; frontend/src/app/api/gmail/messages/[id]/route.ts

3 Persistence allocation

3.1 SQLite

SQLite contains three manually defined tables: users, message_priorities, and priority_events. The database opens lazily on first use at data/mailflow.db relative to the frontend working directory, unless MAILFLOW_DB_PATH is set. Initialization enables WAL journaling and foreign-key enforcement. The repository ignores the database and journal files.

The local model retains information that Gmail labels alone do not provide: the decision source, rule explanation, boolean signals, and a change history. Message content is fetched from Gmail when needed and is not duplicated in SQLite.

3.2 External and transient data

Gmail Authoritative store for message text, headers, labels, read/starred/draft/trash state, and the MailFlow priority labels applied to messages. A Gmail message ID is an external reference in SQLite, not a local foreign key.

Process memory Google OAuth credentials are held in a Map keyed by gmail_session. A process restart discards them. The browser cookie contains the session ID, not the credentials.

Browser memory Live view state and all /demo state are held in React component state. The demo has no persistence or external mutation path.

Implementation: frontend/src/lib/mailflow-db.ts; frontend/src/lib/gmail-session.ts; frontend/src/app/demo/demo-client.tsx; .gitignore

4 Relational schema

The schema is defined by CREATE TABLE statements in frontend/src/lib/mailflow-db.ts. Types and constraints below reproduce the implemented DDL. TEXT timestamps are written with Date.toISOString().

4.1 users

One row identifies one Gmail account in the local database.

id INTEGER PRIMARY KEY AUTOINCREMENT; local account key.

email TEXT NOT NULL UNIQUE; trimmed, lowercased Gmail address.

display_name TEXT NULL; profile name or fallback supplied by the profile route.

created_at TEXT NOT NULL; UTC creation timestamp.

updated_at TEXT NOT NULL; UTC timestamp of the latest upsert.

4.2 message_priorities

One current classification for each (user_id, gmail_message_id) pair. The composite pair is the primary key.

user_id INTEGER NOT NULL; FK to users.id ON DELETE CASCADE.

gmail_message_id TEXT NOT NULL; external Gmail message ID.

priority TEXT NOT NULL; CHECK value in ('urgent', 'fyi', 'low').

source TEXT NOT NULL; CHECK value in ('automatic', 'manual', 'gmail_label').

reason TEXT NOT NULL; explanation associated with the current decision.

signals_json TEXT NULL; JSON object with automated, directAction, timeSensitive, and activeBlocker booleans when automatic classification supplies them.

classified_at TEXT NOT NULL; UTC timestamp of first insert, retained on later priority updates.

updated_at TEXT NOT NULL; UTC timestamp of the latest changed classification.

last_seen_at TEXT NOT NULL; UTC timestamp of the latest recorded observation.

4.3 priority_events

One row per first classification or later change of priority. The application inserts rows and does not update existing events.

id INTEGER PRIMARY KEY AUTOINCREMENT; event key.

user_id INTEGER NOT NULL; FK to users.id ON DELETE CASCADE.

gmail_message_id TEXT NOT NULL; external Gmail message ID.

previous_priority TEXT NULL; CHECK value in ('urgent', 'fyi', 'low'); NULL for first classification.

new_priority TEXT NOT NULL; CHECK value in ('urgent', 'fyi', 'low').

source TEXT NOT NULL; CHECK value in ('automatic', 'manual', 'gmail_label').

reason TEXT NOT NULL; explanation at the time of the event.

signals_json TEXT NULL; JSON-encoded automatic-rule signals when supplied.

created_at TEXT NOT NULL; UTC event timestamp.

Index: priority_events_message_idx on (user_id, gmail_message_id, created_at). Each child table has a many-to-one relationship with users. There is no local messages table.

Implementation: frontend/src/lib/mailflow-db.ts

5 Processing and integrity rules

5.1 Priority assignment

The list route ensures the three hidden Gmail priority labels exist, obtains the account email, and upserts the local user. Daily Briefing lists INBOX messages after the supplied timestamp, traverses Gmail pages, fetches each full message, and filters against internalDate. Ordinary mailbox views fetch one Gmail page at a time.

For a non-draft message, an existing MailFlow label takes precedence. If none appears in the full message, the server requests minimal message metadata once more to detect a label added by another request. Otherwise it evaluates rule signals from the sender, subject, and snippet. A non-automated blocker or time-sensitive direct request is urgent; an automated message without direct action is low; other messages are fyi. The chosen Gmail label is added before the local priority is recorded. Draft messages receive fyi in the response and skip this classification path.

Daily Briefing sorts by priority (urgent, fyi, low) and then by parsed date descending. Ordinary mailbox responses preserve the Gmail list order.

Implementation: frontend/src/app/api/gmail/messages/route.ts; frontend/src/lib/gmail-priority.ts; frontend/src/lib/mail-priority.ts

5.2 Local write behavior

upsertUser normalizes the email and updates display_name only when a non-empty value is supplied. recordPriority compares the incoming priority with the current row. If unchanged, it updates only last_seen_at and does not insert an event. If new or changed, BEGIN IMMEDIATE covers the current-row upsert and event insert; the code commits both or rolls back on error. A manual priority change first updates Gmail labels, then records source=manual locally.

Because Gmail and SQLite are separate stores, the code does not provide a cross-system transaction. A Gmail mutation can succeed before the local record is written. Subsequent message loads can reconcile the current category from the Gmail label.

Implementation: frontend/src/lib/mailflow-db.ts; frontend/src/lib/gmail-priority.ts

5.3 Daily Briefing sequence

The companion sequence diagram traces the valid-session, unlabeled-message path from the browser request through Gmail listing, message retrieval, priority assignment, and SQLite recording. It is supplemental to the relational schema. Source: Documents/MailFlowAI-DailyBriefing-SequenceDiagram.mmd. Rendered copies: .png and .pdf with the same base name.

6 Web-view data traceability

The assignment preview is implemented at /demo. It uses five fixed Mail records and a fixed event list. The following fields exist only in component state unless separately handled by the live Gmail client.

6.1 Preview data structures

Mail id, sender, email, subject, time, preview, body, summary, priority, unread, and optional date and attachment. These values are declared in initialMail.

Priority Urgent, FYI, or Low Priority. The preview uses these display labels; the live API uses urgent, fyi, and low.

Calendar event day, time, title, and place in a fixed events array. Event-review fields additionally include date, time zone, and location.

Transient controls Current view, selected message, filter, query, modal, toast, reply draft, compose recipient/subject/body, tone, length, and event-review fields.

6.2 Views and output fields

Daily Briefing Groups the sample Mail records by priority and shows sender, time, subject, and summary.

All Emails and detail The list shows search/filter results, unread state, preview, and priority. Detail shows original sample body, optional attachment name/date, summary, and editable reply.

Compose and review Recipient, subject, body, summary preview, tone, and length are local state. The summary preview takes the first 24 words of compose text. Tone and length sliders are illustrative. Confirming send displays a demo notice; no email is sent.

Calendar and event review The month grid and event data are fixed preview values. Confirming an event displays a demo notice; no Google Calendar event is created.

The live root client separately loads Gmail-backed Daily Briefing, All Emails, Starred, Drafts, Trash, and message detail. It does not implement the preview's AI summary, compose/send, or calendar-event mutation as live services.

Implementation: frontend/src/app/demo/demo-client.tsx; frontend/src/app/mail-client.tsx

7 Team contributions

The following attribution is based on commits present in the shared Git history. It describes repository work products rather than inferred division of effort.

Donghyun Lim

609335e Initial Gmail-connected Next.js prototype with message loading, session helpers, API routes, and mail client.

31c338a All Emails and Daily Briefing views and Gmail message actions.

291d550 All Emails loading order and pagination changes with tests.

fdfa4c8 SQLite classification schema, local database inspection, and persistence integration.

8ffeb9c Data Design PDF committed on origin/main.

Seunghoon Jang

516322e Calendar view, HTML email viewer, and MIME parsing.

8d32e02 Google OAuth login and callback flow, recovery behavior, documentation, and route tests.

ac67cfe Priority rules, Gmail labels, ordering, manual changes, and related tests.